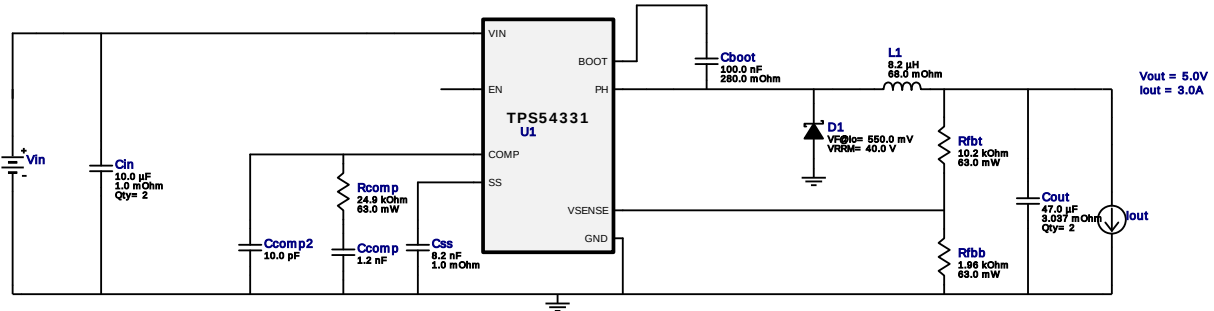


WEBENCH® Design Report

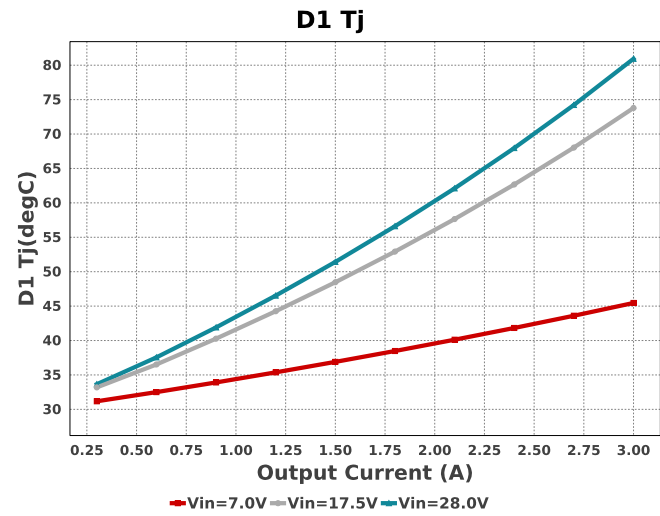
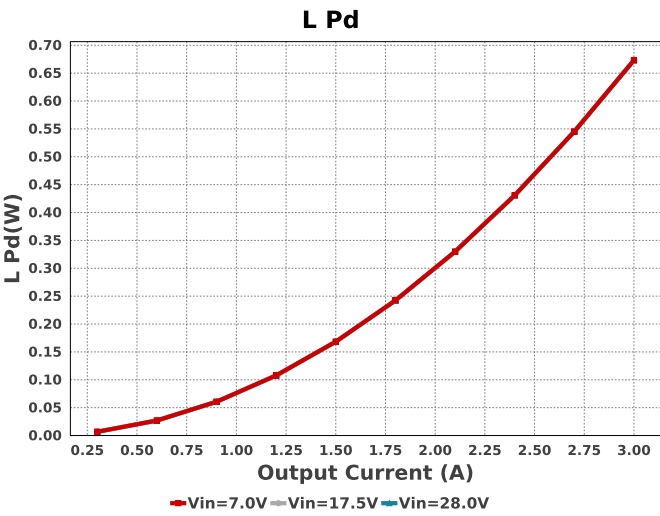
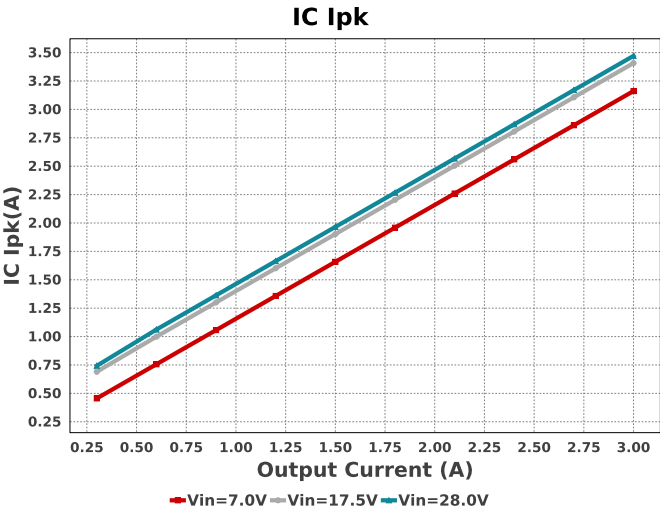
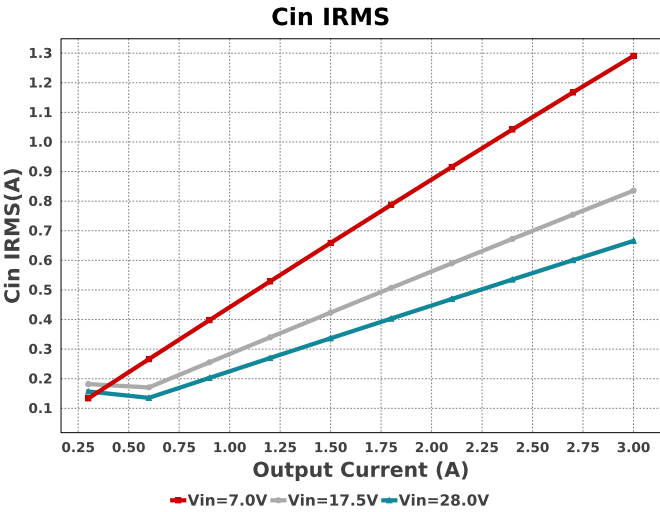
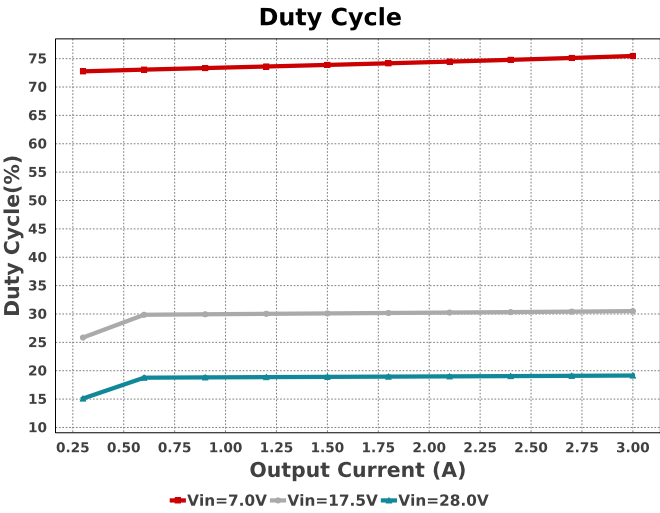
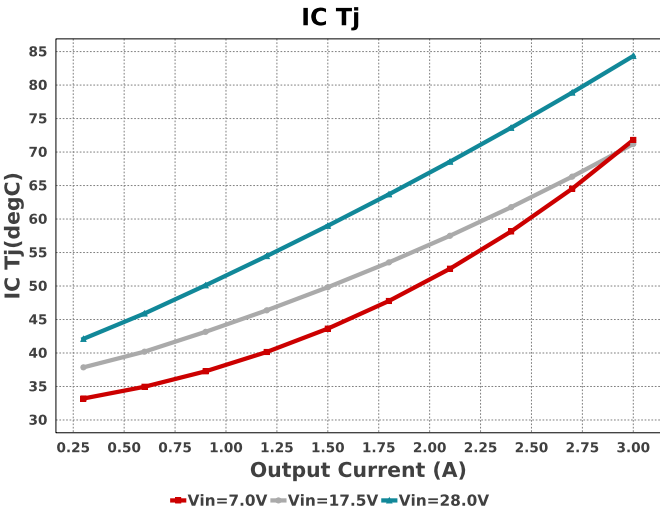
Design : 1 TPS54331DDAR
TPS54331DDAR 7V-28V to 5.00V @ 3A

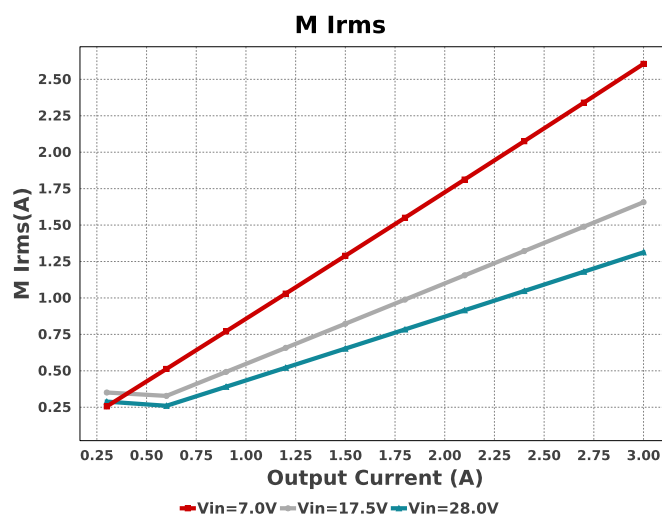
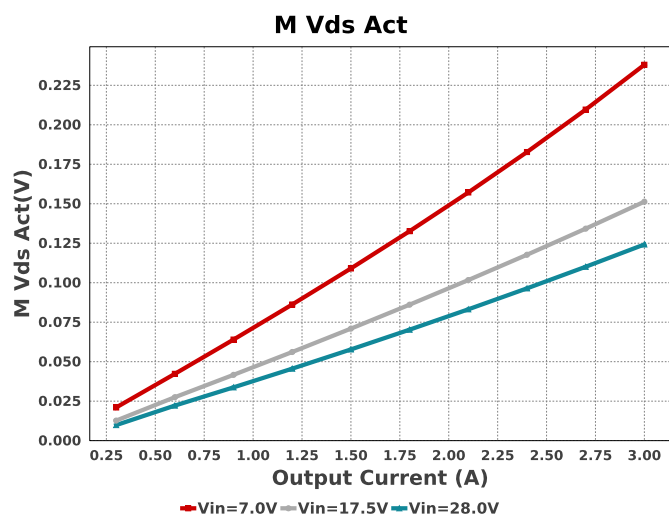
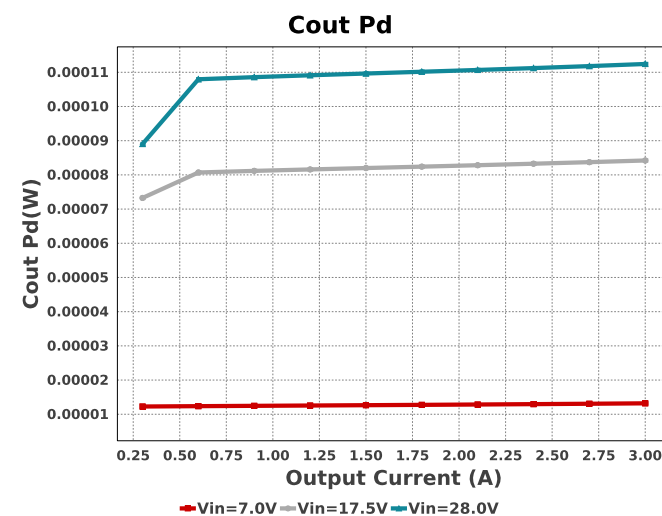
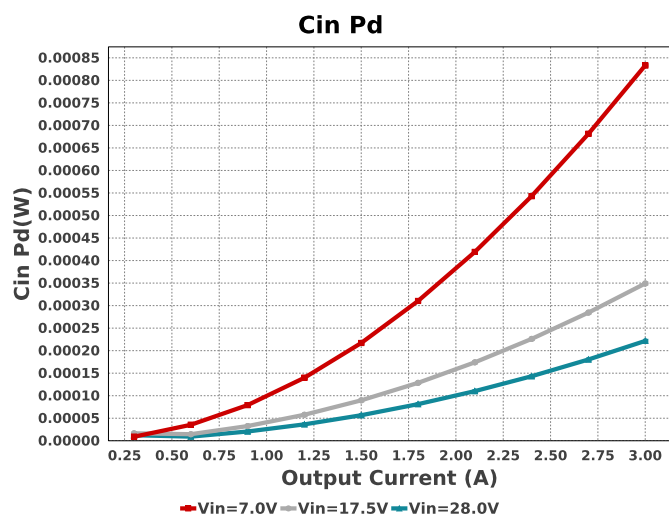
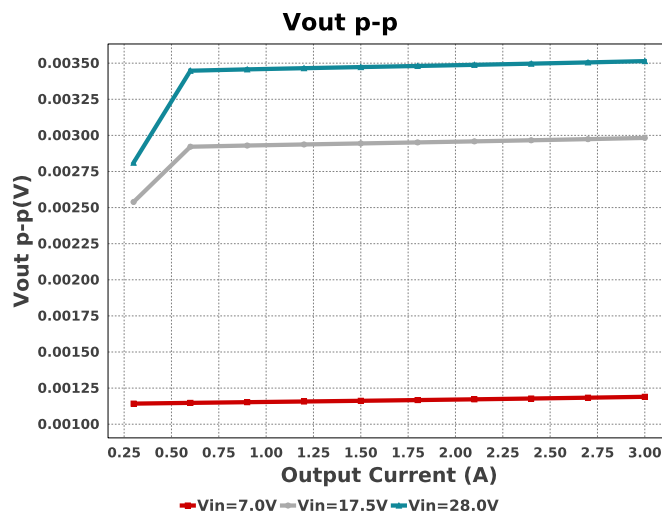
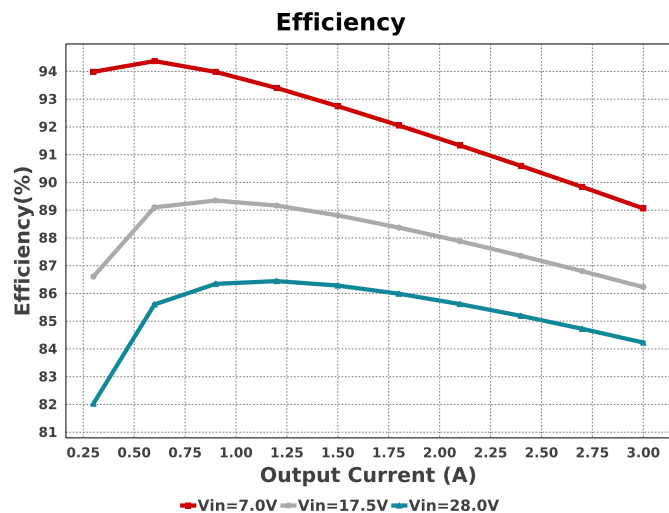


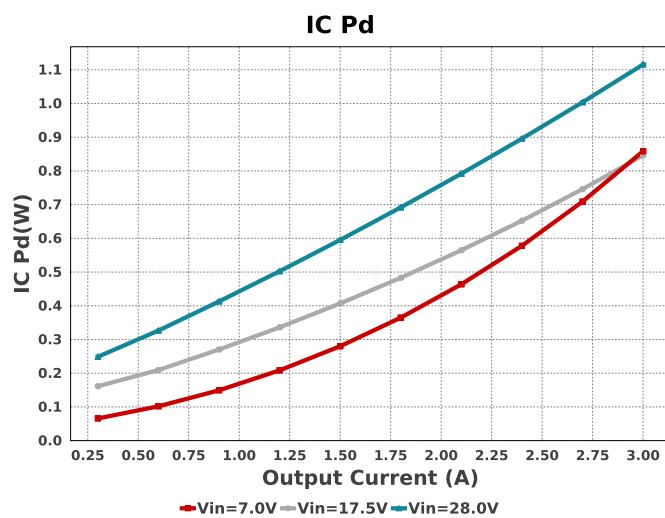
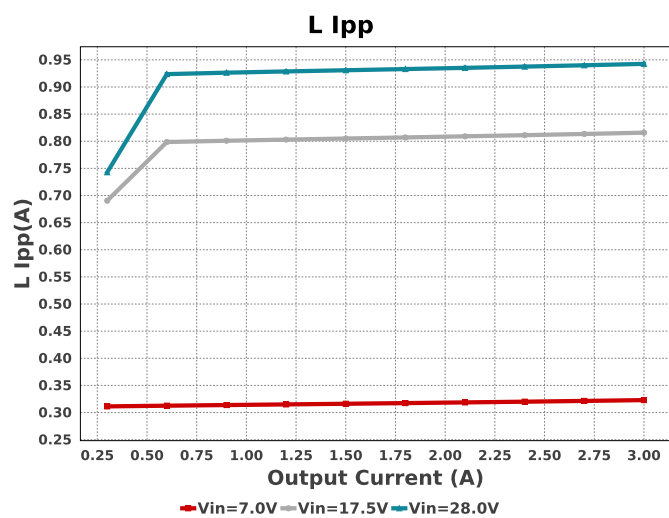
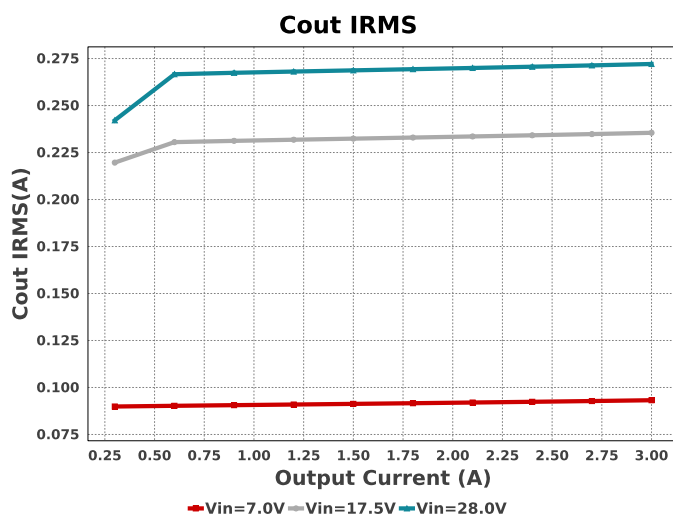
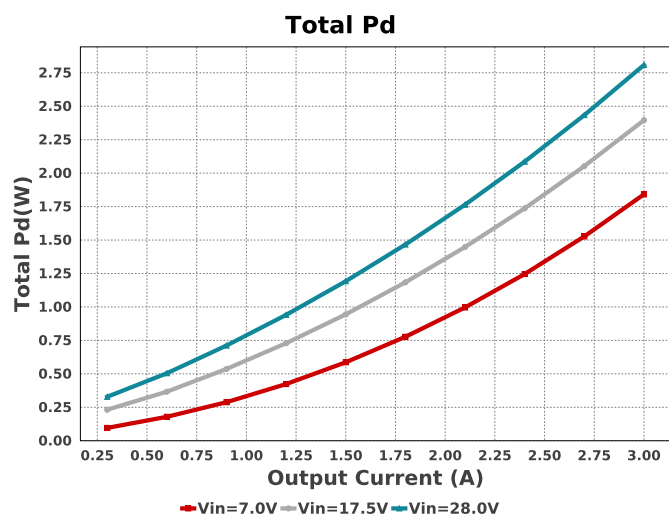
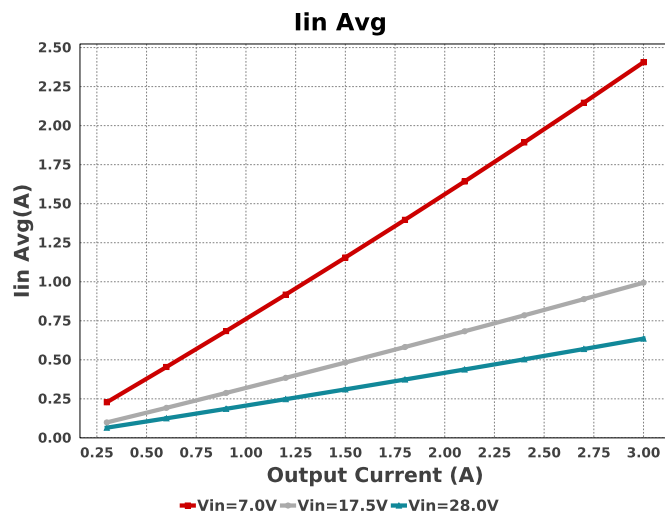
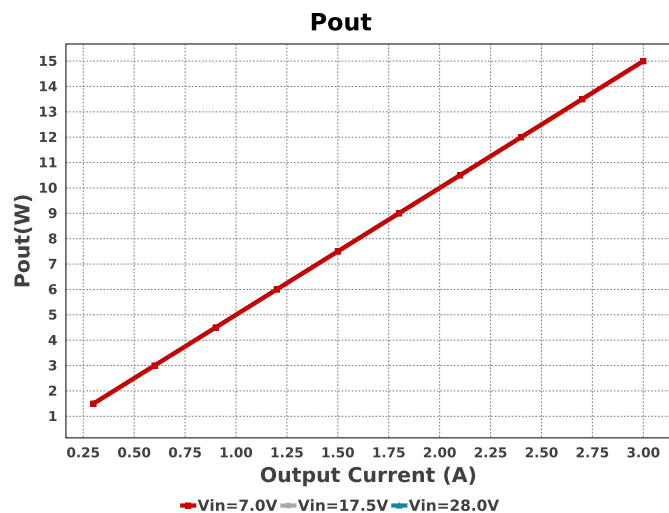
Electrical BOM

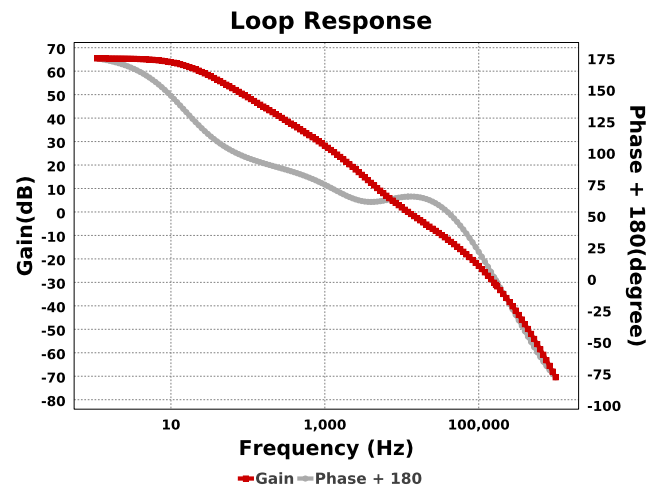
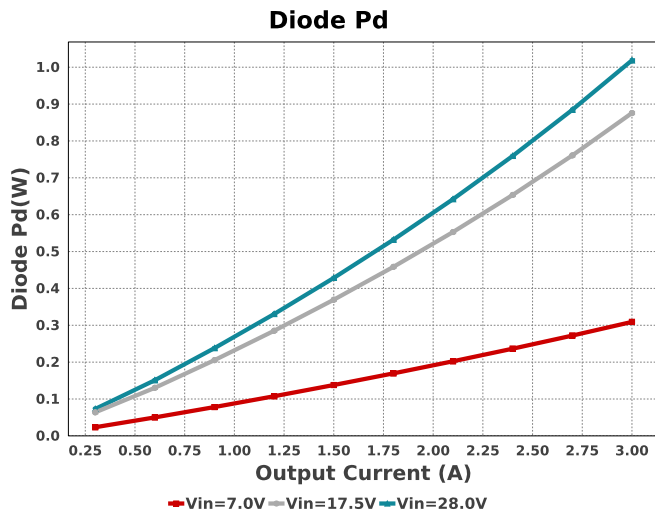
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	\$0.01	0805 7 mm ²
Ccomp	Kemet	C0805C122J5GACTU Series= C0G/NP0	Cap= 1.2 nF VDC= 50.0 V IRMS= 0.0 A	1	\$0.03	0805 7 mm ²
Ccomp2	Yageo	CC0805JRNPO9BN100 Series= C0G/NP0	Cap= 10.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	0805 7 mm ²
Cin	TDK	C3225X7R1H106M250AC Series= X7R	Cap= 10.0 uF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 5.0 A	2	\$0.27	1210 15 mm ²
Cout	MuRata	GRM32ER61C476KE15L Series= X5R	Cap= 47.0 uF ESR= 3.037 mOhm VDC= 16.0 V IRMS= 4.59346 A	2	\$0.17	1210_280 15 mm ²
Css	MuRata	GRM033R71A822KA01D Series= X7R	Cap= 8.2 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	\$0.01	0201 2 mm ²
D1	Diodes Inc.	B540C-13-F	VF@Io= 550.0 mV VRRM= 40.0 V	1	\$0.19	SMC 83 mm ²
L1	Vishay-Dale	IHLP2525CZER8R2M01	L= 8.2 uH 68.0 mOhm	1	\$0.61	IHLP-2525CZ 75 mm ²
Rcomp	Vishay-Dale	CRCW040224K9FKED Series= CRCW..e3	Res= 24.9 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfbb	Vishay-Dale	CRCW04021K96FKED Series= CRCW..e3	Res= 1.96 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
Rfbt	Vishay-Dale	CRCW040210K2FKED Series= CRCW..e3	Res= 10.2 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
U1	Texas Instruments	TPS54331DDAR	Switcher	1	\$0.71	 DDA0008H 55 mm²









Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	665.755 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	221.62 μ W	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	272.1 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	112.43 μ W	Capacitor	Output capacitor power dissipation
5.	D1 Tj	80.937 degC	Diode	D1 junction temperature
6.	Diode Pd	1.019 W	Diode	Diode power dissipation
7.	IC Ipk	3.471 A	IC	Peak switch current in IC
8.	IC Pd	1.116 W	IC	IC power dissipation
9.	IC Tj	84.34 degC	IC	IC junction temperature
10.	ICThetaJA	48.7 degC/W	IC	IC junction-to-ambient thermal resistance
11.	Iin Avg	636.0 mA	IC	Average input current
12.	L Ipp	942.58 mA	Inductor	Peak-to-peak inductor ripple current
13.	L Pd	673.2 mW	Inductor	Inductor power dissipation
14.	M Irms	1.313 A	Mosfet	MOSFET RMS ripple current
15.	M Vds Act	124.228 mV	Mosfet	Voltage drop across the MosFET
16.	Cin Pd	221.62 μ W	Power	Input capacitor power dissipation
17.	Cout Pd	112.43 μ W	Power	Output capacitor power dissipation
18.	Diode Pd	1.019 W	Power	Diode power dissipation
19.	IC Pd	1.116 W	Power	IC power dissipation
20.	L Pd	673.2 mW	Power	Inductor power dissipation
21.	Total Pd	2.808 W	Power	Total Power Dissipation
22.	BOM Count	14	System	Total Design BOM count
23.	Cross Freq	12.068 kHz	System	Bode plot crossover frequency
24.	Duty Cycle	19.155 %	System	Duty cycle
25.	Efficiency	84.231 %	System	Steady state efficiency
26.	FootPrint	304.0 mm ²	System	Total Foot Print Area of BOM components
27.	Frequency	570.0 kHz	System	Switching frequency
28.	Gain Marg	-31.058 dB	System	Bode Plot Gain Margin
29.	Iout	3.0 A	System	Iout operating point
30.	Low Freq Gain	65.42 dB	System	Gain at 1Hz
31.	Mode	CCM	System	Conduction Mode
32.	Phase Marg	66.237 deg	System	Bode Plot Phase Margin
33.	Pout	15.0 W	System	Total output power
34.	Total BOM	\$2.48	System	Total BOM Cost
35.	Vin	28.0 V	System	Vin operating point
36.	Vout	5.0 V	System	Operational Output Voltage

#	Name	Value	Category	Description
37.	Vout Actual	4.963 V	System Information	Vout Actual calculated based on selected voltage divider resistors
38.	Vout Tolerance	5.254 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
39.	Vout p-p	3.514 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	3.0	Maximum Output Current
VinMax	28.0	Maximum input voltage
VinMin	7.0	Minimum input voltage
Vout	5.0	Output Voltage
base_pn	TPS54331	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

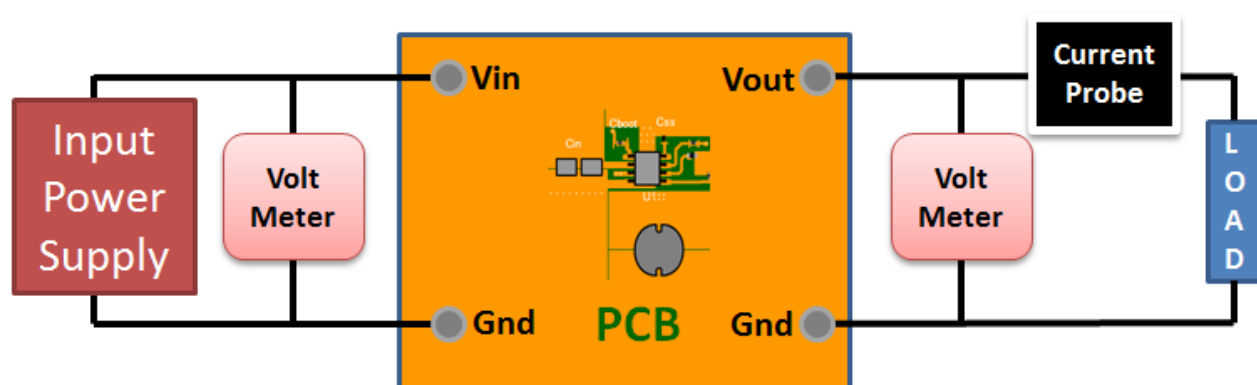
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 7.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

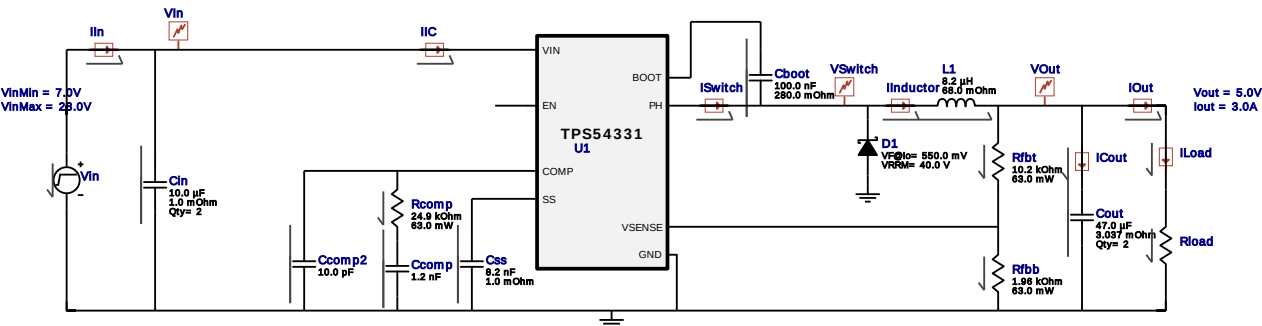
Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



WEBENCH® Electrical Simulation Report

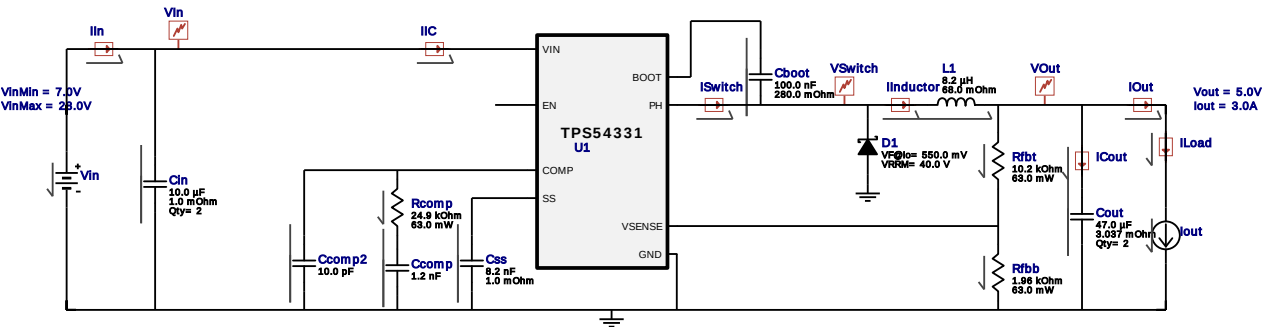
Design Id = 1
sim_id = 1
Simulation Type = Startup



Simulation Parameters

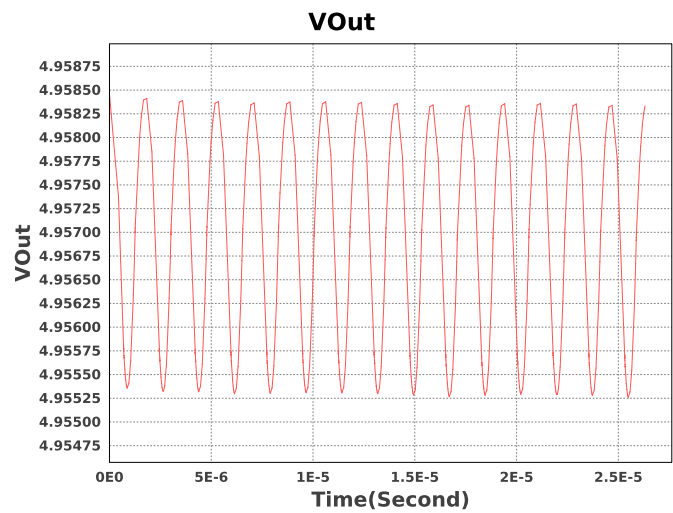
#	Name	Parameter Name	Description	Values
1.	Rload	R	Load Resistance	1.666666666666667 Ohm

Design Id = 1
sim_id = 2
Simulation Type = Steady State



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Css	IC	Initial Voltage	1 V
2.	Cboot	IC	Initial Voltage	17.5 V
3.	L1	IC	Initial Current	3.0 A
4.	Iout	I	Load Current	3.0 A



Design Assistance

- 1. Master key : C93D2D7EACCB66BA80879407777D2B7[v1]
- 2. **TPS54331** Product Folder : <http://www.ti.com/product/TPS54331> : contains the data sheet and other resources.

Important Notice and Disclaimer

TI provides technical and reliability data (including datasheets), design resources (including reference designs), application or other design advice, web tools, safety information, and other resources AS IS and with all faults, and disclaims all warranties. These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, or other requirements.

These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you will fully indemnify TI and its representatives against, any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

Providing these resources does not expand or otherwise alter TI's applicable Terms of Sale or other applicable terms available either on [ti.com](https://www.ti.com) or provided in conjunction with TI products.